STE 80858-1

TCRseries
Industrial Robot

TCR10LA/TS3100

INSTRUCTION MANUAL

TRANSPORTATION & INSTALLATION

Notice

1. Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
2. Before installing and operating the industrial robot, read through and completely understand this manual.
3. After reading through this manual, keep it nearby for future reference.

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Shibaura Machine

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PREFACE

This manual describes how to unpack and install the robot and robot controller. Specifically, it describes how to unpack the shipments containing the robot and controller, how to install them, how to connect wiring and air piping and how to mount a tool on the robot arm. Be sure to read through this manual before unpacking each shipment.

Before beginning the work according to this manual, read through the Setup Manual and Safety Manual so that you can learn the names and functions of various components and the safety measures.

This manual is divided into the following four (4) sections.

Section 1: Transportation
This section describes how to unpack the shipments containing the robot and controller and how to carry them to the installation site. Also, it describes the precautions to be observed when the equipment are to be stored temporarily after the unpacking.

Section 2: Installation
This section discusses the environment in which the robot and controller are installed, space requirements and how to install the equipment.

Section 3: System Connections
This section describes how to connect the robot, controller and peripheral equipment.

Section 4: Hand Interface
This section describes how to mount a tool on the robot arm and how to connect wiring and air piping to the tool.

CAUTIONS ON SAFETY

This manual contains the important information on the robot and controller to prevent injury to the operators and persons nearby, to prevent damage to assets and to assure correct use.
Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
 **DANGER**	This means that "incorrect handling will lead to fatalities or serious injuries."
 **CAUTION**	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2)."

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.

*2) Physical damage refers to damages due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). Details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory (must be done). Details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger or caution. The details of the actual caution are indicated with pictures or words in or near the symbol.

CAUTION

- Be sure to read through the Safety Manual provided separately before starting actual work to assure safety of the work covering from the robot installation to operation.

[Installation and transportation]

Be sure to observe the following items to use the robot safely.

DANGER	
 Prohibited	• DO NOT install or operate if any parts are damaged or missing. Otherwise, electric shocks, fires or faults could be caused. • DO NOT install the robot where it may be exposed to splash of liquids such as water. Otherwise, electric shocks, fires or faults could be caused. • DO NOT place the robot near the combustibles. Otherwise, fires could be caused if the material ignites due to a fault, etc.
 Mandatory	• Be sure to secure the robot with the attached clamps before carrying the robot. Otherwise, you could be involved in injuries if the arm moves when the robot is lifted. • Connect the robot only after installation. If wiring is performed before the robot installation, you could be involved in electric shocks or injuries. • Be sure to use the master power voltage and power capacity designated by SHIBAURA MACHINE. Otherwise, the equipment could be damaged or catch fires. • Be sure to use the designated power cable. Otherwise, fires or faults could be caused.
 Always ground	• Completely connect the grounding cable. Otherwise, electric shocks or fires could be caused if a fault or leak current occurs, in addition to malfunction by noise.

! CAUTION	
 Prohibited	• NEVER lift the robot arm during the transport. Otherwise, an excessive force will be exerted on the mechanical part of the robot, resulting in faults. • For the controller, keep an ample space for air vent. Otherwise, the controller will be heated, causing faults.
 Mandatory	• When lifting up the robot, do so carefully. If lifted suddenly, the robot will tilt, which is very dangerous. • When storing the robot, completely clamp the base. If placed directly on the floor, the robot will be unstable and fall over.

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Section 1 Transportation

1.1 Unpacking

The robot and controller are shipped separately in wooden crates and corrugated cardboards. The packaging state of the controller is shown in Fig. 1.1.
The teach pendant and other accessories including instruction manuals are packed separately.
Open the shipments at a location easily accessible for subsequent transport and installation. At this time, take careful precautions not to damage the robot and controller. After unpacking, make sure that all accessories are present and that all equipment have not been damaged during transport.

! DANGER

- If any parts of the equipment are found defective or damaged, or if some accessories are missing when the robot and controller have reached your office, DO NOT install or operate the equipment. Otherwise, the equipment could malfunction. Contact us immediately at our sales agent in your territory.
- After unpacking, dispose of the wooden crates, corrugated cardboards, polyethylene shipping bags and cushion materials according to your regulations.

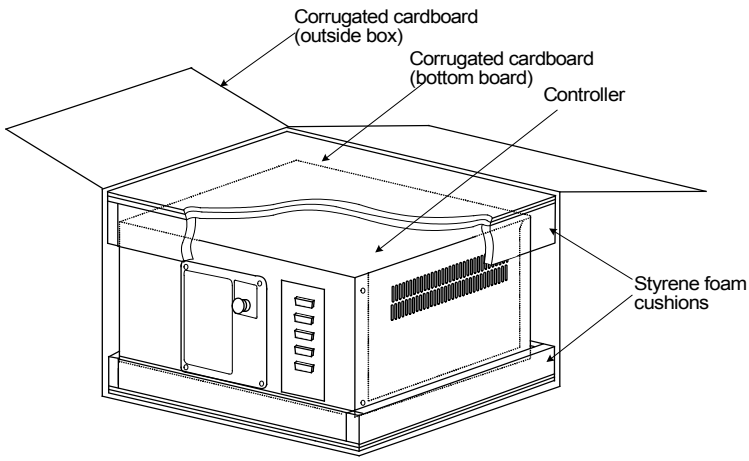


Fig. 1.1 Packaging state of controller

1.2 Transport

Carefully move the robot and controller to avoid excessive impact (9.8 m/s^2 or over) or vibration from being exerted on them. If the equipment are subject to vibration over a long period of time, be sure to completely tighten all nuts and screws.
When relocating the equipment at another place, apply the cushions as originally set, then put the equipment back into the wooden crates and corrugated cardboards.

1.2.1 Mass and External Dimensions

The mass of controller TS3100 is 17 kg. For the mass of the basic robot, see the relevant specifications.

1.2.2 Transporting the Robot

When transporting the robot, observe the following procedures, fully confirming the cautions.

! DANGER

- When transporting the robot, be sure to use the attached lifting fixtures.
- When transporting the robot, be sure to secure the arm with the attached clamp beforehand. Otherwise, a hazardous situation may be caused because the arm will move when the robot is suspended.

When transporting the robot by a crane, observe the following procedures.

1) Make sure that the arm clamps are attached.

2) Make sure that the eyebolts are attached to the transport fixtures.

3) Pass wire ropes through the eyebolts. To assure safety, be sure to use four (4) wire ropes. If there is a fear that the wire rope will contact the robot body due to the length of the wire rope, use a cushion.

4) Lift up the robot carefully, using the crane.

Next, disconnect the robot body from the transport fixtures to carry the robot.

5) Remove the hexagon socket head cap screws and washers, then mount the M10 eyebolts in the screw holes instead.

6) Remove the arm clamps and move so that the arms can overlap each other.

7) Pass wire ropes through the eyebotls. To assure safety, be sure to use the four (4) wire ropes. If there is a fear that the wire rope will contact the robot body due to the length of the wire rope, use a cushion.

8) After the installation, remove the eyebolts and mount the attached hexagon socket head cap screws and washers in the screw holes.

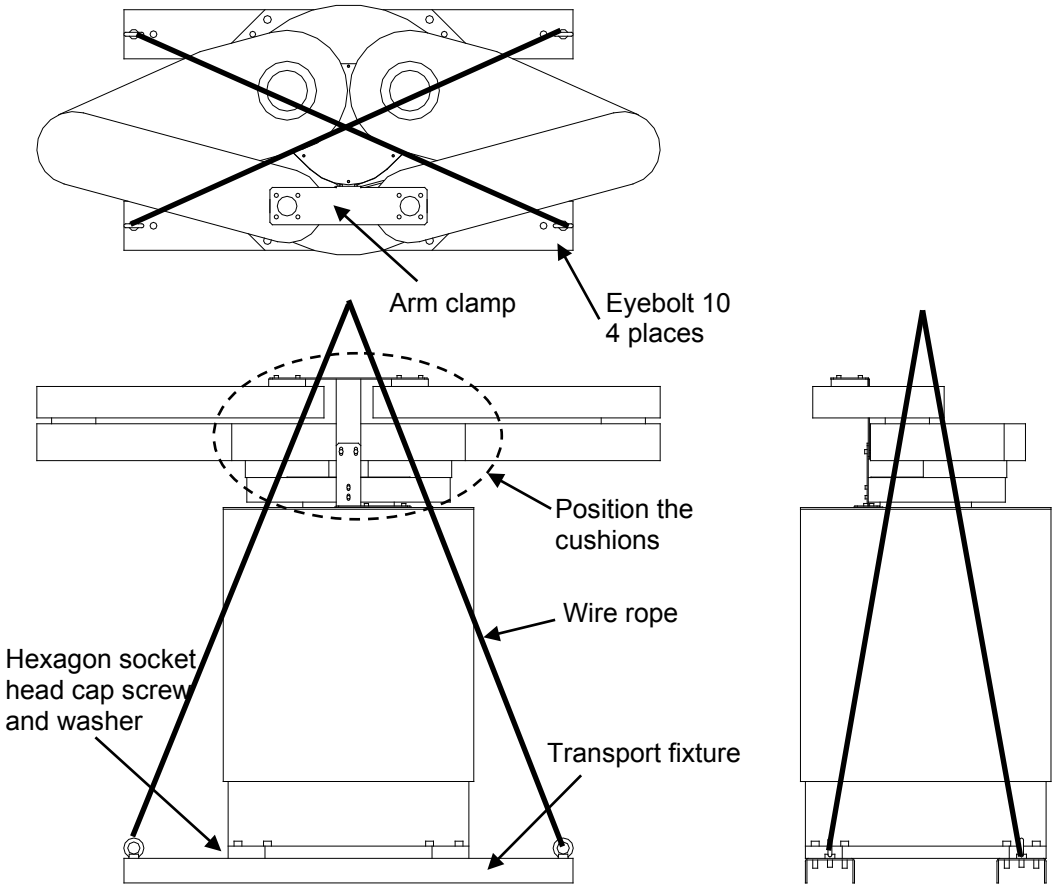


Fig. 1.2 Lifting posture of the robot (double-hand specifications)

! CAUTION

- Use the wires which can well withstand the robot mass.
- The robot will tilt more or less when lifted. Carefully lift it up.
- When raising and lowering the robot, take careful precautions not to exert any excessive impact (9.8 m/s^2 or over) on the robot.
- When placing the robot on the floor, etc., take utmost care not to have your hand or foot caught under the robot.

1.2.3 Transporting the Controller

Before transporting the controller, disconnect all cables and teach pendant.

! DANGER

- When placing the controller on the floor, etc., take utmost care not to have your hand or foot caught under the controller.

1.3 Storage

Both the robot and controller should not be stored for a long period of time after unpacked. When this is unavoidable, however, be sure to observe the following cautions on storage.

1.3.1 Cautions on Robot Storage

! CAUTION

- Secure the robot base completely to prevent the robot from falling over.
- Keep the robot out of direct sunlight. Otherwise, the parts may deteriorate.
- Seal the robot in a vinyl bag to prevent rust development and contaminant. Put a desiccant in the bag to absorb moisture.
- When the robot motor drive cable was disconnected from the controller, connect the short-circuit connector to the same cable.

1.3.2 Cautions on Robot Controller Storage

! CAUTION

- Keep the robot controller out of direct sunlight. Otherwise, the controller interior will be heated up, resulting in faults.

Section 2 Installation

2.1 Installation Environment

Table 2.1 shows the environmental conditions for the location in which the robot and controller TS3100 are to be installed.

Table 2.1 Environmental conditions for the robot and controller

Item	Specifications
Temperature	At operation : 0 ~ 40°C At storage : –10°C ~ 50°C
Humidity	20 ~ 90 % (non-condensing) DO NOT install the robot and controller at places where they are exposed to splash of liquids such as water.
Altitude	1,000 m or less
Vibration	At operation : 0.98 m/s^2 or less At transport : 9.8 m/s^2 or less
Dust	No inductive dust should be involved. Consult with SHIBAURA MACHINE first if you have to use the robot and controller in a hostile environment.
Gas	No corrosive gas or combustible gas should be involved.
Sunlight	The robot and controller should not be exposed to direct sunlight.
Power source noise	No device causing an excessive noise should exist nearby.
Magnetic field	No device causing a strong magnetic field should exist nearby.

! DANGER

- DO NOT place the robot and controller near combustibles. Otherwise, fires could be caused if the material ignites due to a fault, etc.

2.2 Installing the Robot

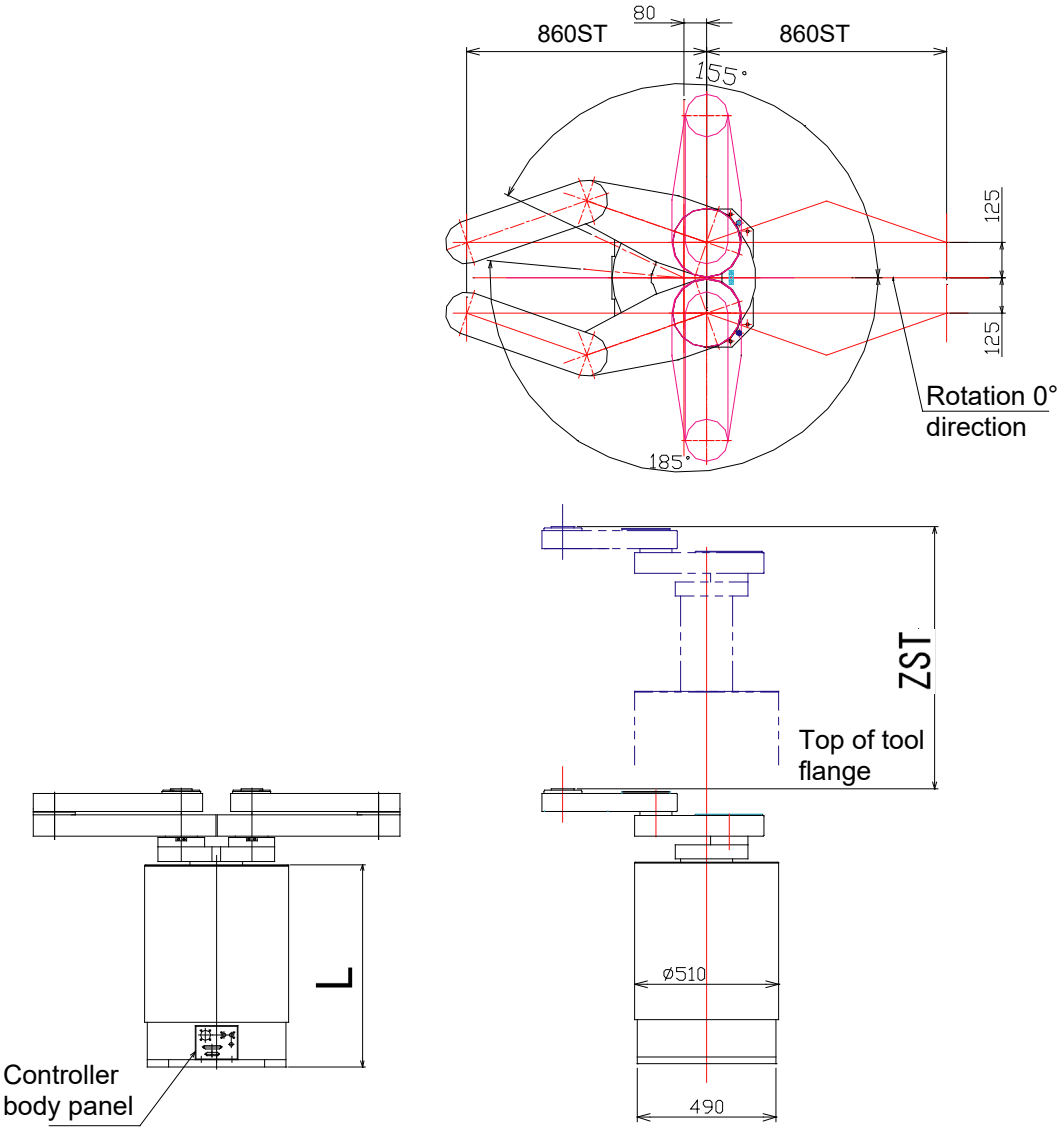
Before actually installing the robot, you should review a layout, fully considering the working envelope, coordinate system and space for maintenance.

2.2.1 External Dimensions

See the external view drawing of the robot attached to the specifications.

2.2.2 Working Envelope

See the external view drawing of the robot attached to the specifications.



ZST(mm)	780	930
L(mm)	645	715

Fig. 2.1 External view

2.2.3 Origin Posture and Coordinate System of the Robot

The origin posture and coordinate system of the robot are exemplified in Fig. 2.2.

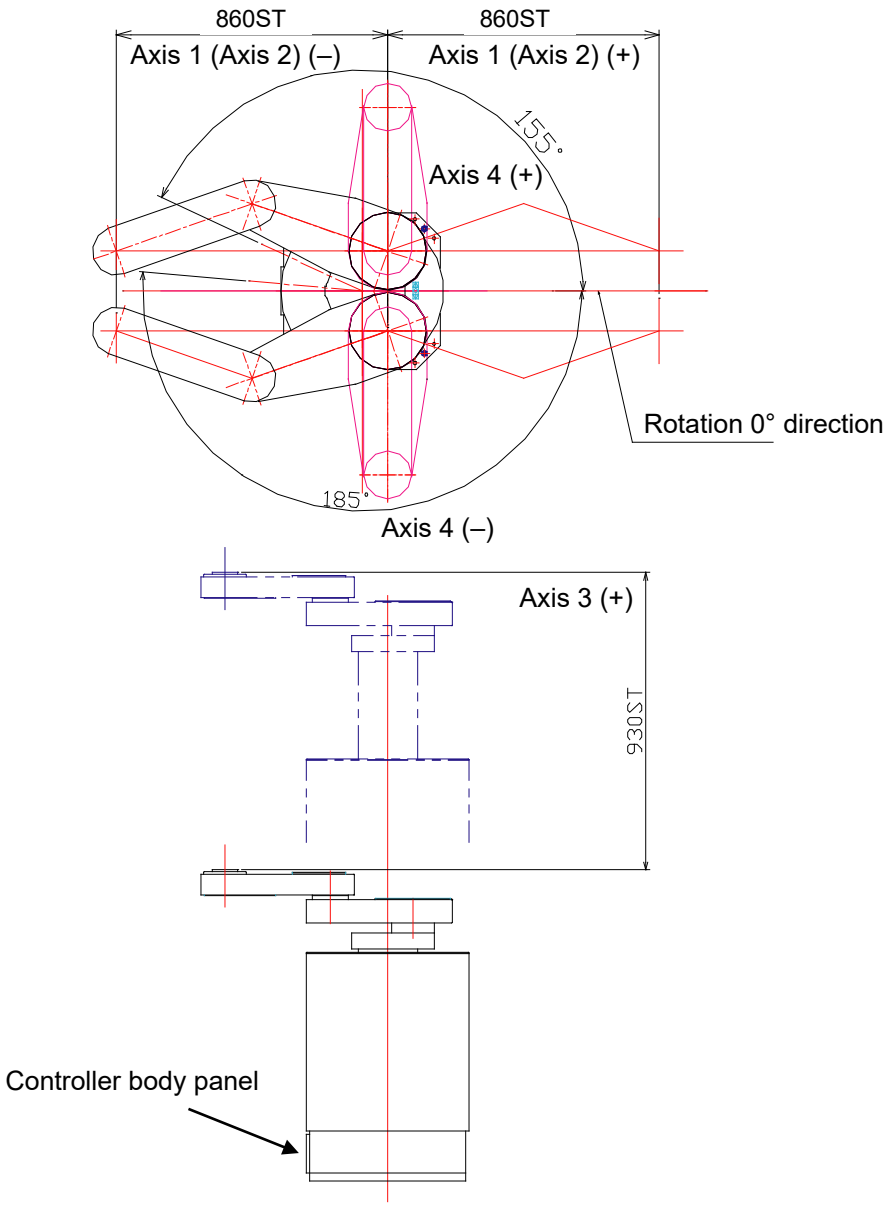


Fig. 2.2 Coordinate system of the robot

2.2.4 Robot Installation Procedures

The robot is secured, using the set holes on the robot base.
The robot base set areas should be rigid enough and their surfaces should be machined with flatness accuracy as small as 0.2 or less.

! CAUTION

- The robot will suddenly accelerate and decelerate during operation. When installing the robot on the frame, make sure that the frame has enough rigidity. If the robot is installed on a frame that is not rigid enough, vibration will be caused during the robot operation, resulting in faults.
  When installing the robot on the floor, secure it with anchor bolts, etc.
- Install the robot on the leveled place. Otherwise, the robot performance will deteriorate or the robot will go wrong.

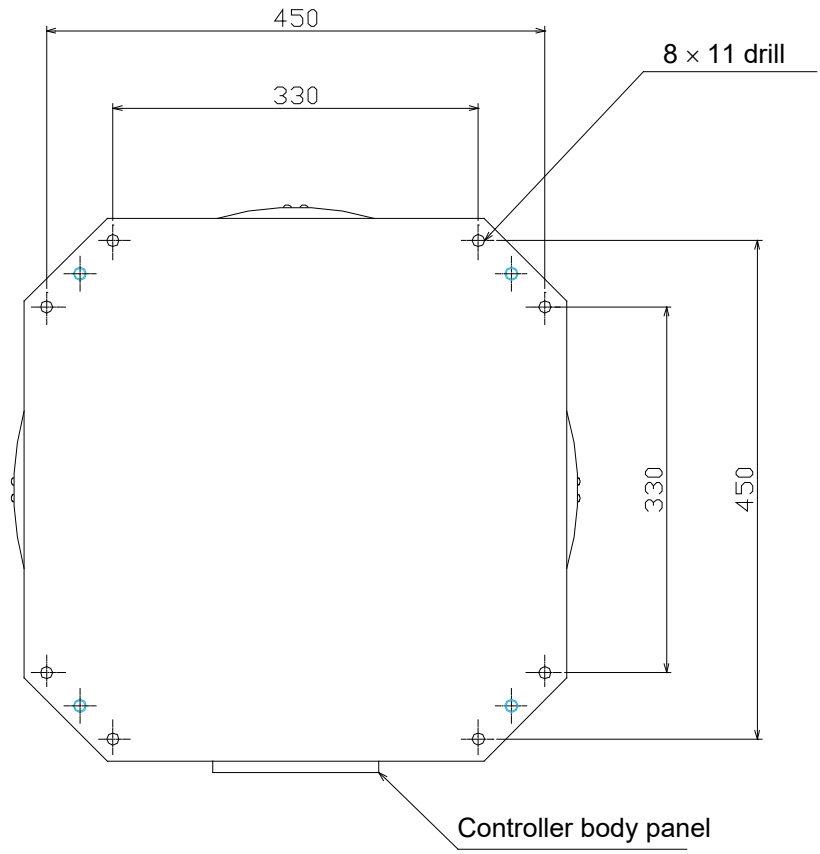


Fig. 2.3 Robot base

2.3 Installing the Controller

2.3.1 External Dimensions

Fig. 2.4 shows the external dimensions of the robot controller.

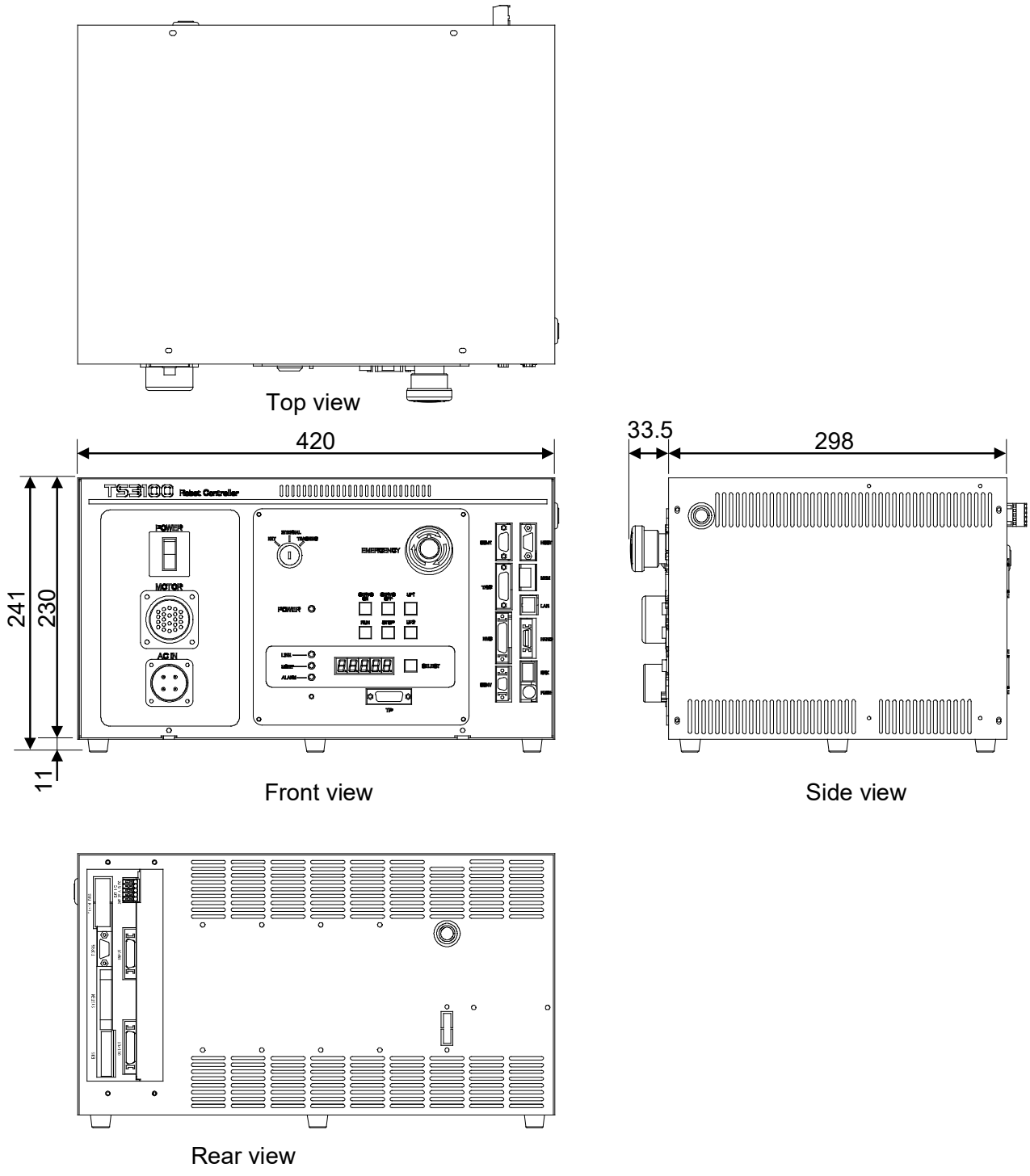


Fig. 2.4 External view of TS3100 controller

2.3.2 Rack Mounting Dimensions

When installing the TS3100 robot controller in the rack, set the side brackets, using the screw holes on both ends of the front panel, then secure the controller to the rack. (Side brackets [1] in Fig. 2.5 are optional.)

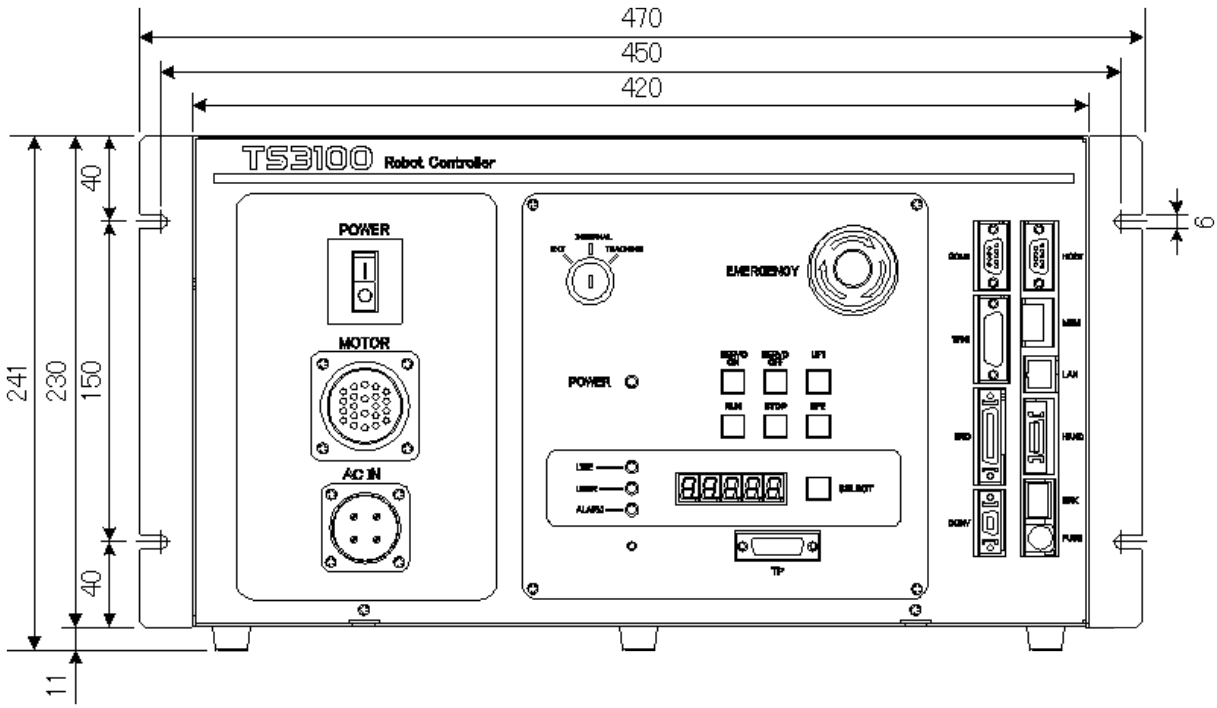


Fig. 2.5 Dimensions of controller rack clamp screw holes

2.3.3 Cautions on Rack Mounting

Take careful precautions on the following matters when mounting the TS3100 robot controller in the rack.

a) When mounting the controller in the rack, use the screw holes provided on both ends of the front panel and secure the controller. (The optional side brackets are required.)

! CAUTION

- If the rack is completely sealed, provide holes for air vent, or compulsively ventilate the rack with a fan or cool the rack indirectly, so that the heat will not be trapped in the rack.
  If the heat is trapped in the rack or controller, the controller will malfunction.

b) As the cable connectors are connected to the rear side of the controller, keep a space of 110 mm or over on the rear side. To perform maintenance of the controller, the upper cover is removed. (See Fig. 2.6.)
When installing the controller, fully consider the controller maintenance. When the controller is housed in the rack, it should be removed from the rack at maintenance. Specifically, take careful precautions on the following points.

1) Arrange the cables around the rear side of the controller (so that the controller can be removed).

2) Arrange the cables between the controller and control panel when the control panel is separated (so that the controller can be removed).

3) Connect all cables in a position where the robot can be operated even if the controller is removed from the rack.

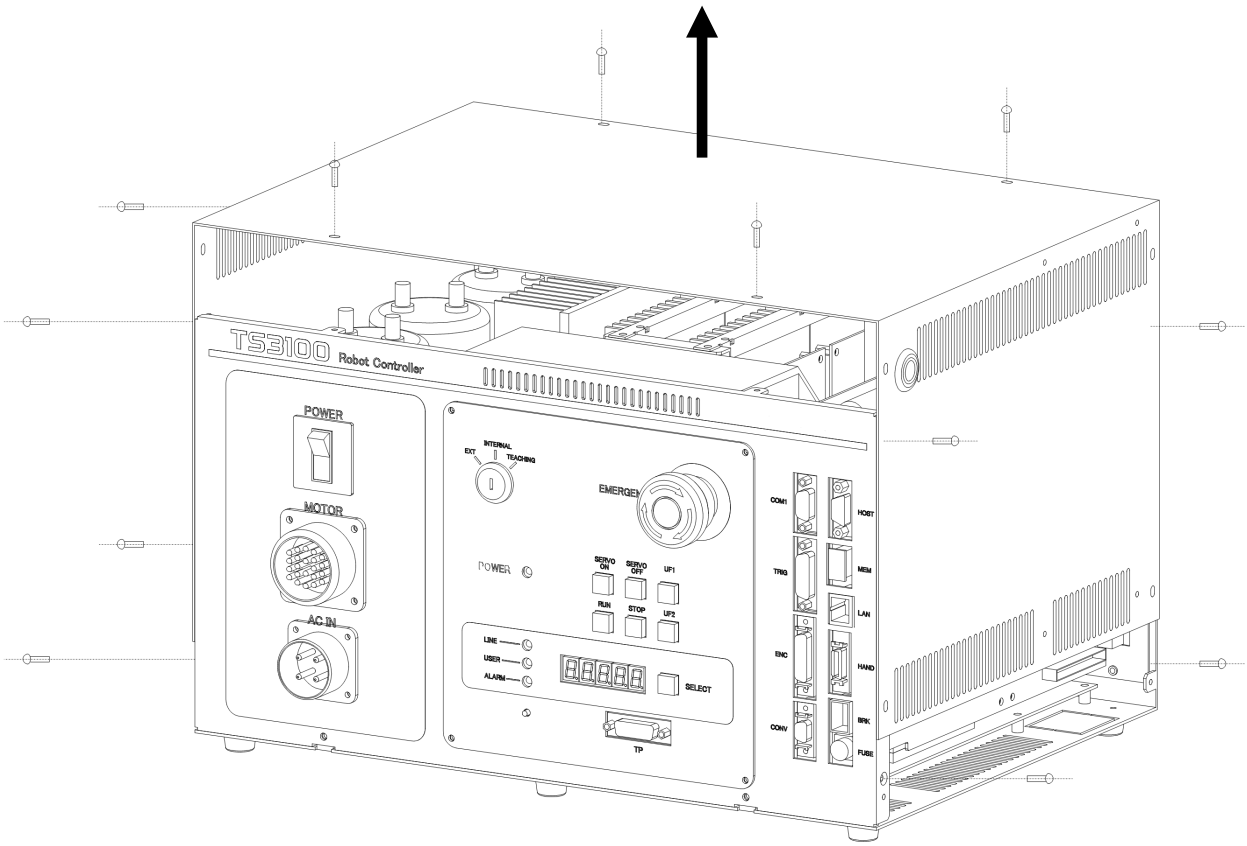


Fig. 2.6 Removing upper cover

c) When the controller is assembled in the rack, the weight of the controller should be supported by the legs of the controller. The rack mount screw holes of the controller are used for securing the controller panel, and the weight of the controller cannot be supported only with these screws.

d) On the front side of the controller, a clearance of about 90 mm should be kept for connecting the connector of the teach pendant. Even if the teach pendant is not used, a clearance of about 60 mm is required for connecting a dummy plug.

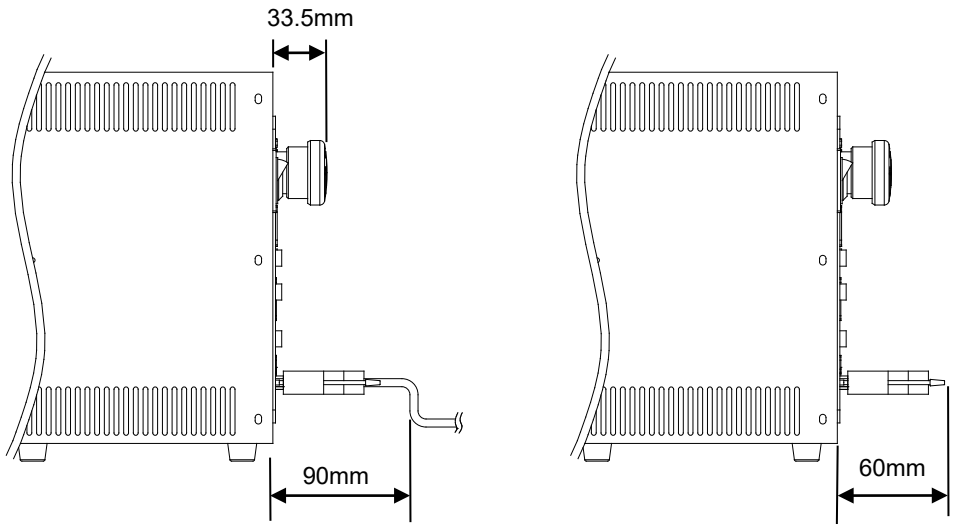


Fig. 2.7 Clearance on controller front side

2.3.4 Cautions on Direct Installation

When two (2) or more controllers are used, keep a clearance of 50 mm or over in the right and left directions between the controllers and a clearance of 100 mm or over in the upper direction of each controller.

! CAUTION

- Keep a ventilation space on the lateral side of the controller to prevent stoppage of the air vent holes. Also, keep the space equal to the length of the leg below the bottom surface. Otherwise, the cooling ability will drop, causing faults.
- DO NOT stack the controllers. DO NOT put objects on the top of each controller.

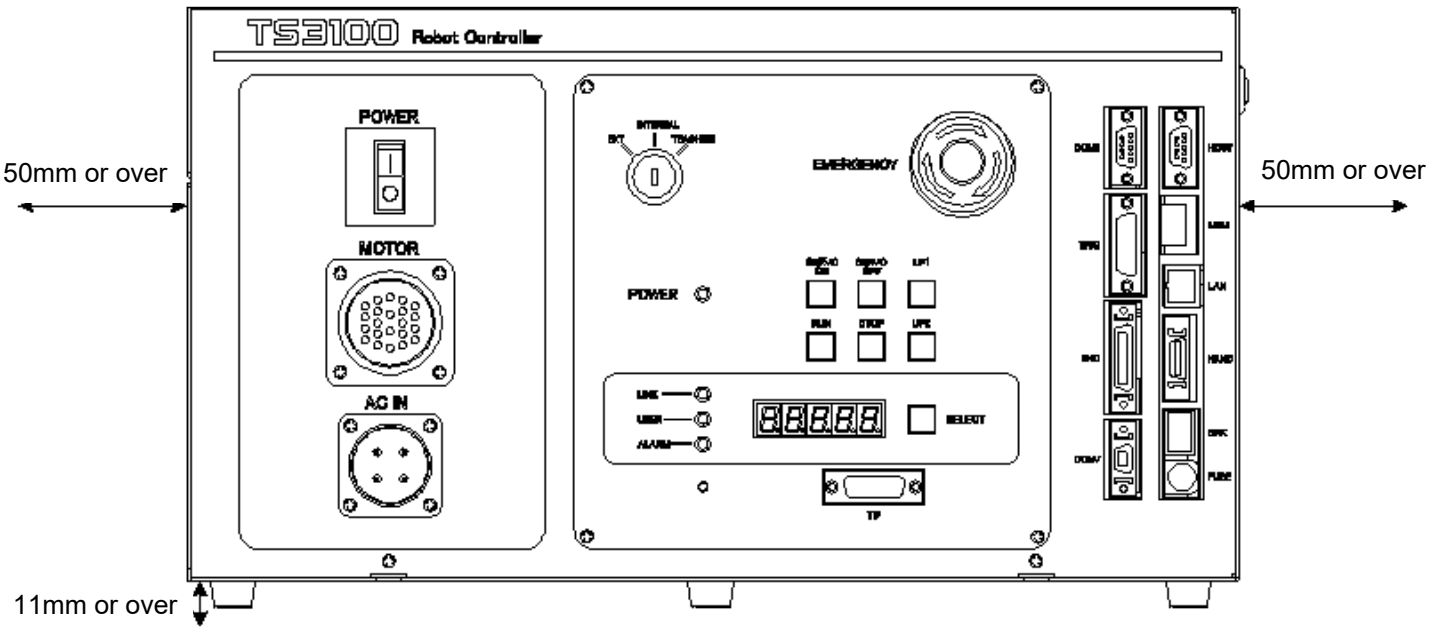


Fig. 2.8 Controller ventilation space

2.4 Cautions on Handling of Teach Pendant

When handling the teach pendant, observe the following precautions.

! CAUTION

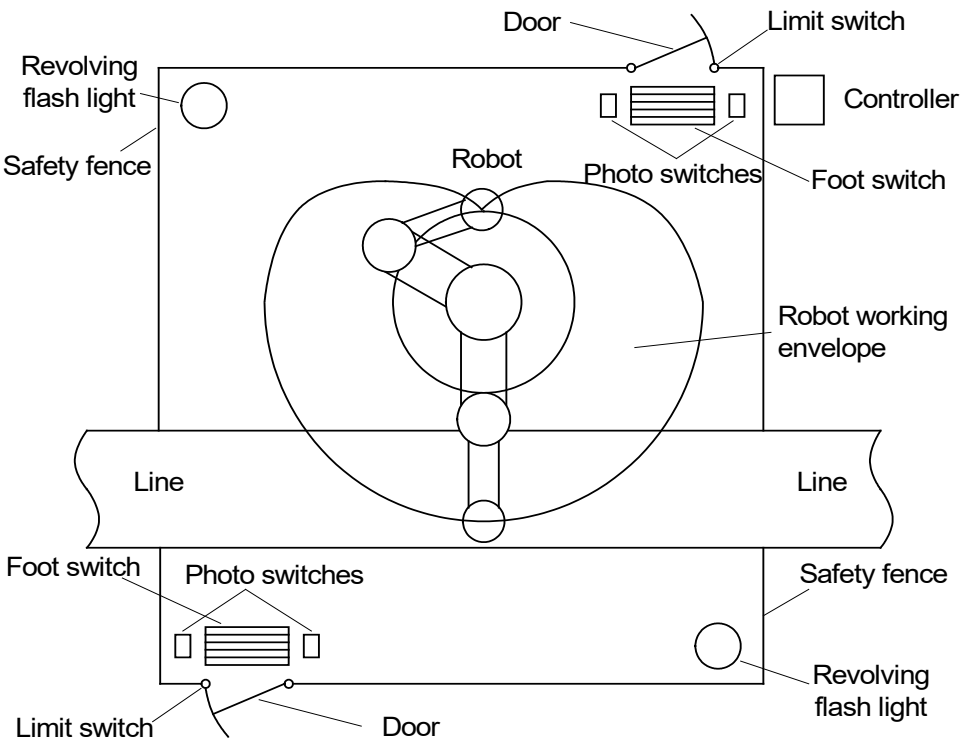
- DO NOT drop the teach pendant or hit it against anything.
- DO NOT pull the cable coming out of the teach pendant.
- DO NOT press the switches on the teach pendant with a sharp cutter, pencil or ball-point pen.
- DO NOT place or use the teach pendant near open flames.
- DO NOT leave the teach pendant in direct sunlight for a long period of time.

2.5 Safety Measures

a) When installing the robot, provide sufficient space to carry out the work safely.

b) Clarify the hazard zone, and provide safety fences to prevent anyone from entering the zone easily. The hazard zone is the zone near the working envelope of the robot, where a hazardous state could occur if a person enters.

c) Mount the limit switches, photo switches, foot switches, etc., on the entrance/exit doors of the safety fences and provide an emergency stop function that will stop the robot if a person enters the hazard zone. The emergency stop function should be an electrically independent normal close contact (close during normal operation) with compulsive opening function and should not be recovered automatically.



d) The controller should be installed at a place outside the hazard zone where the operator can overlook the robot motions.

Section 3 System Connection

3.1 Cable Connection

This section describes the various types of cables and connectors and explains how these are to be connected.

3.1.1 Connector Layout of the Controller

The cables connected to the TS3100 robot controller are shown in Fig. 3.1

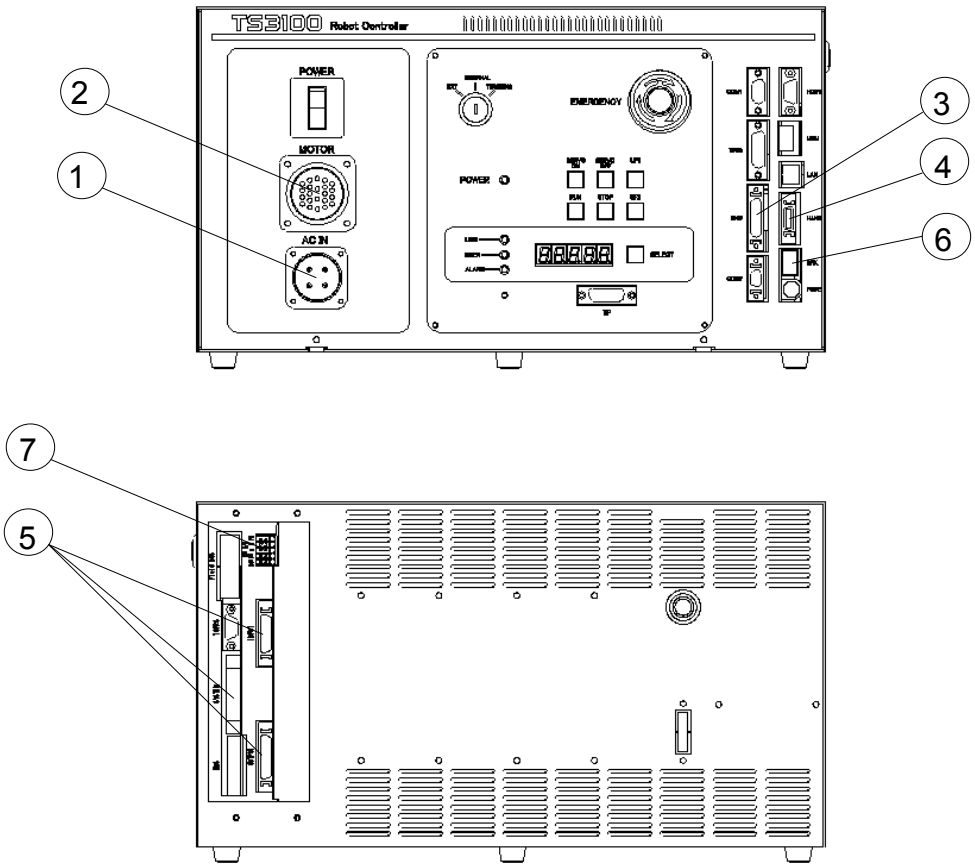


Fig. 3.1 Robot controller connector arrangement

① Power cable (ACIN)
② Motor cable (MOTOR)
③ Encoder cable (ENC)
④ Robot control signal cable (HAND)
⑤ External operation I/O signal cables (SYSTEM,INPUT,OUTPUT)
⑥ Brake signal cable (BRK)
⑦ Distribution I/O signal cable (EXT-I/O)

In the subsequent paragraphs, we explain how to connect cables ① to ④ inclusive. For information on how to connect cables ⑤, ⑥ and ⑦, refer to the Interface Manual.

3.1.2 Connecting Power Cable " ACIN " (① of Fig. 4.1; plug connector attached)

The power cable is used to supply the main power to the controller.

Table 3.1 Power supply specifications

Item	Specifications
Power supply	Single-phase, AC 200 ~ 240 V, 50/60 Hz±1 Hz
Power capacity	2.7 kVA
Instantaneous power failure	Within 40 msec
Grounding	Class D grounding

The connector is ACIN (① of Fig. 3.1)

ACIN1 plug connector	Type: JL04V–6A22-22SE-EB-R	Maker:	Japan Aviation Electronics Industry
CN1 cable clamp	Type: JL04–2022CK(14)-R	Maker:	Japan Aviation Electronics Industry
Wire	3.5 mm^2 ~ 5.5 mm^2		

As the cable is not an accessory, use the attached plug connector connected to ACIN on the controller side to manufacture a cable.
Wires are to be soldered to the connector.

! DANGER

- Be sure to use the designated wires. Otherwise, fires or faults could be caused.
- When connecting the connector and wires, make sure of the terminal arrangement.
- After the connection, confirm the connection, using a tester, etc.

For the terminal arrangement, see Para. 3.1.7.

CAUTION

- Unless the main power is normally supplied to the controller due to phase defect or voltage drop, "8–027 Slow Charge error" occurs.
  When this happens, make sure that the master power voltage at the controller power connector satisfies the specified input power of the controller, and that the same master power voltage is stabilized.
- For details of the 8–027 error, see the Operation Manual.

3.1.3 Connecting Motor Cable "MOTOR" (② of Fig. 3.1) (Cable attached)

The motor cable connects the controller and robot, and supplies the power required to rotate the motor from the controller servo driver to each axis feed motor of the robot. The motor cable is connected to the connector "MOTOR" (② in Fig. 3.1). The motor cable on the robot side is connected to the connector "BN2" (① in Fig. 3.2).

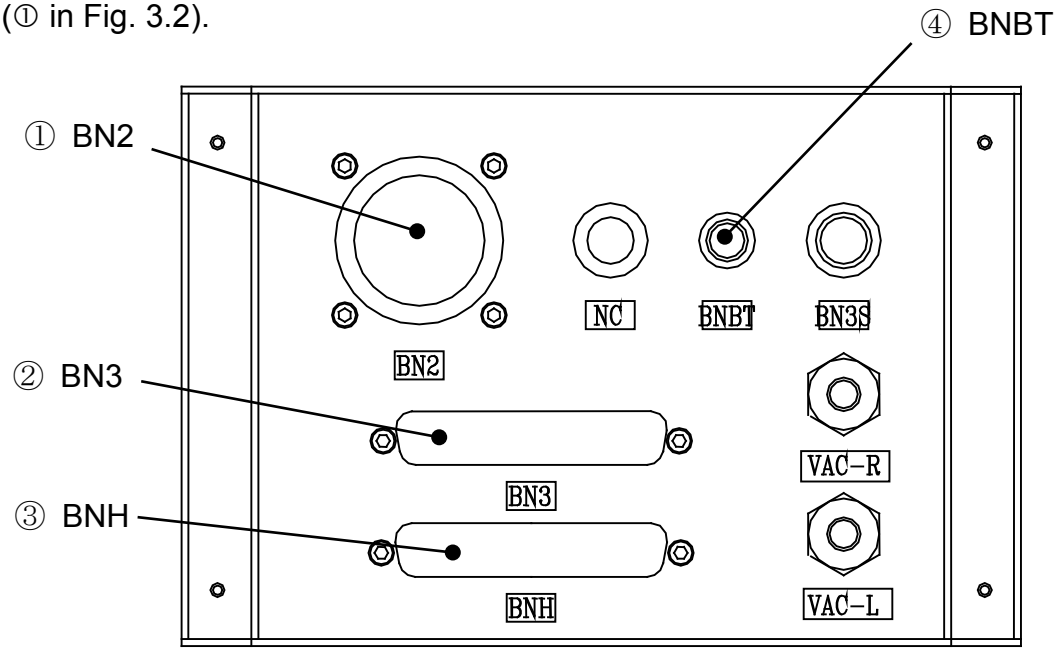


Fig. 3.2 Robot side connector arrangement

CAUTION

- DO NOT pull out the connector ”BNBT” (④ in Fig. 3.2) because it is connected with the Internal battery.

3.1.4 Connecting Encoder Cable "ENC" (③ of Fig. 4.1) (Cable attached)

The encoder cable is a signal line used to transmit a signal from the rotation angle detecting encoder of each robot axis to the controller.
The connector for connecting the encoder cable is “ENC” (③ of Fig. 3.1).
The encoder cable on the robot side is connected to the connector “BN3” (② in Fig. 3.2).

3.1.5 Connecting Robot Control Signal Cable "HAND" (④ of Fig. 3.1) (Cable: option)

The robot control signal cable is used for motor brake ON and OFF, and input and output of robot control signals such as hand operation signal.
The robot control signal cable is connected to the connector "HAND" (④ of Fig. 3.1).
The robot control signal cable on the robot side is connected to the connector "BNH" (③ of Fig. 3.2).

3.1.6 Connecting and Disconnecting Cables

! CAUTION

- Before connecting or disconnecting any controller cable, be sure to turn off the main power (“POWER”) switch.
- When disconnecting a cable, be sure to pull the plug and not the cord. Otherwise, you may damage the cable.

a) Circular connectors: ACIN, MOTOR
Firstly align the key position, and completely insert the connector on the cable side into the controller connector. Then turn the cable side lock screw to the right to clamp the cable. A loose screw can cause a contact failure or other accident. To avoid this, make sure that the screw is clamped completely.
To disconnect the connectors, turn the lock screw to the left and pull out the cable side connector.

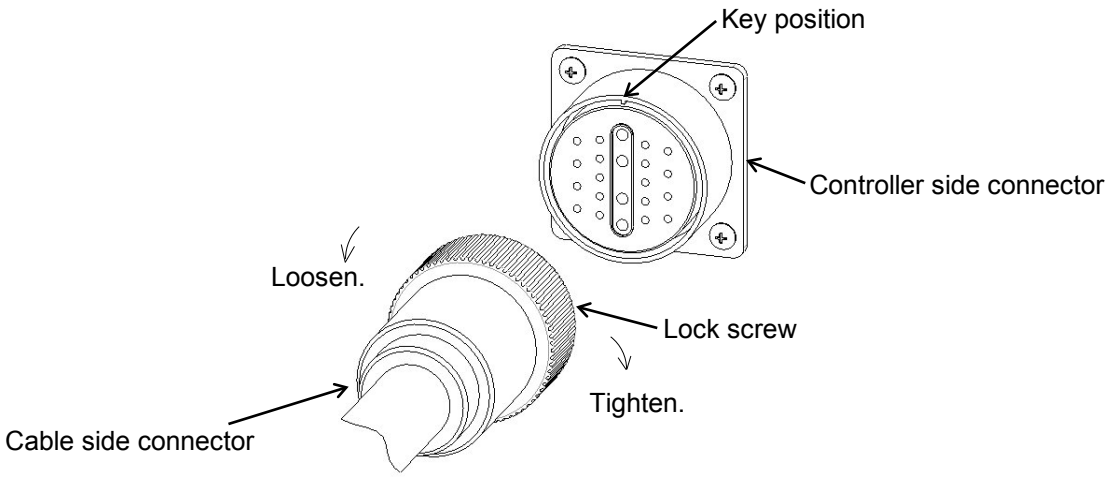


Fig. 3.3 Connecting and disconnecting a circular connector

b) Square connectors: ENC, HAND, SYSTEM, INPUT, OUTPUT, TRIG, CONV
Firstly, completely insert the cable side connector into the controller connector. Then tighten the lock screws on both ends of the cable side connector with a screwdriver. A loose screw can cause a contact failure or other accident. To avoid this, make sure that the screws are clamped completely
To disconnect the connectors, first loosen the lock screws, then pull out the cable side connector. INPUT and OUTPUT are quick-operated lock type connectors, however.

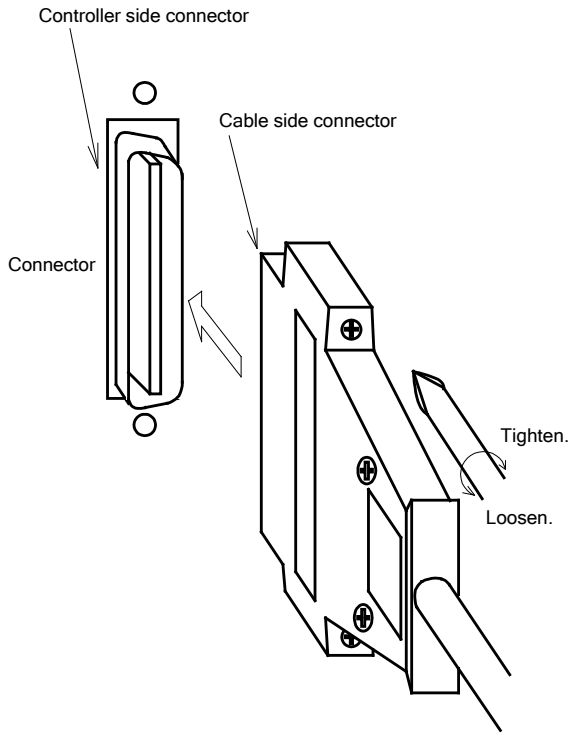
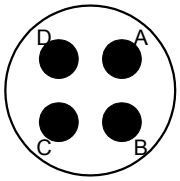


Fig. 3.4 Connecting and disconnecting a square connector

3.1.7 Examples of Connector Terminal Arrangement

a) Power cable connector ACIN



Connects to controller.

Type: JL04HV-2E22-22PE-B

Manufacturer:
Japan Aviation
Electronics Industry

A
B
C
D

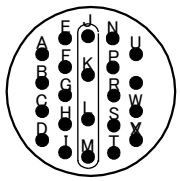
Single-phase, AC 200~240 V, 50/60 Hz

Grounding (Class D or higher level)

DANGER

- Completely connect the grounding cable. Otherwise, an electric shock or fire may be caused if a fault or electric leak occurred. Or mis-operation may be caused by noise.

b) Motor cable connector MOTOR

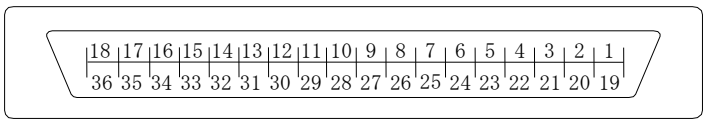


Connects to controller.

Type: JL04V–2A28–11SE

Manufacturer: Japan Aviation Electronics Industry

c) Encoder cable connector ENC

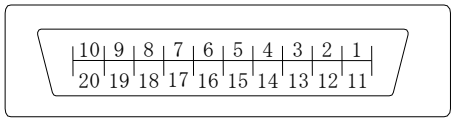


Connects to controller.

Type: 52986-3659

Manufacturer: Molex

d) Robot control signal cable connector HAND

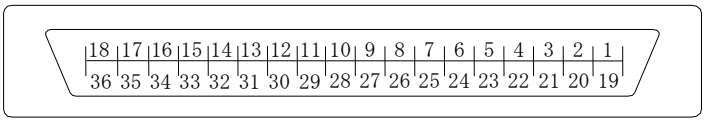


Connects to controller.

Type: 52986-2079

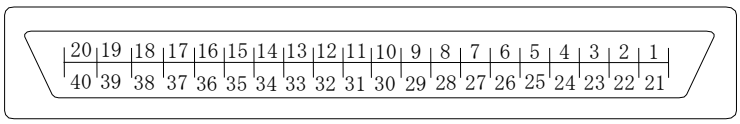
Manufacturer: Molex

e) General purpose input signal cable connector INPUT



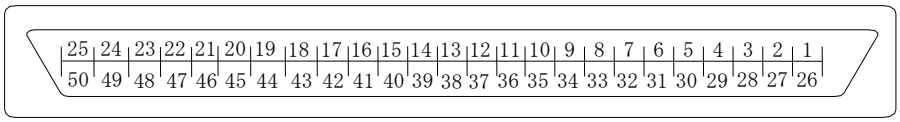
Connects to controller.
Type: DHA-RC36-R132N-FA
Manufacturer: DDK Ltd.

f) General purpose output signal cable connector OUTPUT



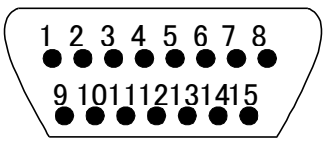
Connects to controller.
Type: DHA-RC40-R132N-FA
Manufacturer: DDK Ltd.

g) System input/output signal cable connector SYSTEM



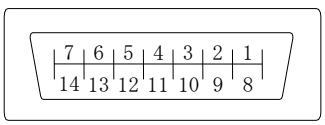
Connects to controller.
Type: 52986-5079
Manufacturer: Molex

h) Trigger input connector TRIG



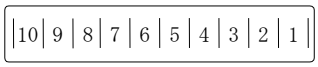
Connects to controller.
Type: XM2C-1542-112L
Manufacturer: Omron

i) Encoder cable connector CONV



Connects to controller.
Type: 52986-1479
Manufacturer: Molex

j) Emergency stop, safety input, and external 24 V power connector EMS

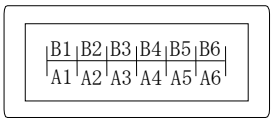


Connects to controller.

Type: ML-4000CWJH-10PGY

Manufacturer: Sato Parts

k) Brake connector BRK



Connects to controller.

Type: 1-1827876-6

Manufacturer: Tyco Electronics

3.2 Controller Connector Signals

3.2.1 Connector Signal Connection Diagrams

Diagrams showing which signals correspond to which terminals are shown in Section 2 of the Interface Manual.

3.2.2 Jumpers for Safety Related Signals

The following system input signals are provided to serve for the safety purpose.

System input signals	SYSTEM-12	(STOP)
	SYSTEM-16	(SVOFF)
	SYSTEM-14	(BREAK)
	EMS-7, 8	(EMS2B ~ EMS2C)
	EMS-9, 10	(EMS1B ~ EMS1C)
	EMS-3, 4	(ENA2B ~ ENA2C)
	EMS-5, 6	(ENA1B ~ ENA1C)

These signals are already jumpered for the connectors provided for the TS3100 robot controller. If you wish to use or change them, therefore, you should remove the jumpers and rewire as appropriate. If you plan to use the robot without using system input signals, be sure to connect the attached connectors to the controller side SYSTEM, EMS connectors.
Unless the following signals are used as the system signals, jumper them also.

SYSTEM-15 (LOW_SPD)
SYSTEM-13 (CYCLE)

Connector jumpers

SYSTEM		EMS	
12-17 (18)	14-17 (18)	3-4	5-6
16-17 (18)	(13-17 (18))	7-8	9-10
(15-17 (18))	-		

! CAUTION

- Unless the signals of SVOFF and emergency stop contacts 1, 2 are jumpered, the controller servo power cannot be turned on.
- Unless the CYCLE signal is jumpered, the controller enters the cycle operation mode.
- Unless the LOW_SPD signal is jumpered, the robot is operated at low speed during automatic operation.
- Unless the STOP signal is jumpered, automatic operation of the robot is not possible.
- Unless the BREAK signal is jumpered, automatic operation of the robot is not possible.

3.3 Separating Control Panel from Controller

3.3.1 Removing Control Panel

Remove the control panel in the following manner.

a) Loosen the four (4) screws at the four (4) corners, which secure the control panel.

b) Remove these screws, then carefully draw out the control panel toward your side.

Caution: Be careful of the cable connected on the rear side.

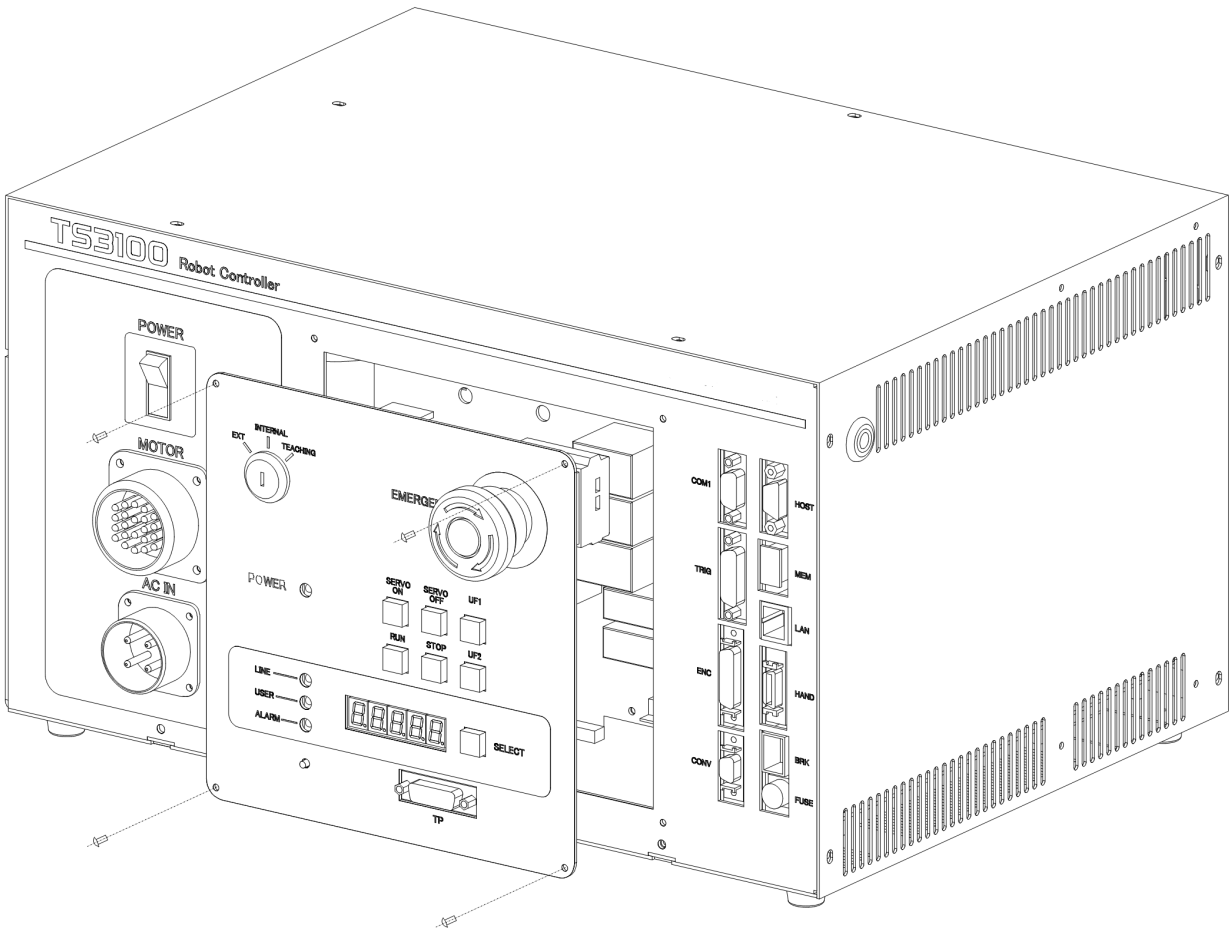


Fig. 3.5 Removing control panel

3.3.2 Cable between Controller and Control Panel

The cable required to connect the control panel and controller when they are installed separately can be provided optionally.

3.3.3 Control Panel Mounting Dimensions

The dimensions of mounting the control panel are shown in Fig. 3.6.
Cross truss head screws (ø3 x 6, ZN3–B) are used.

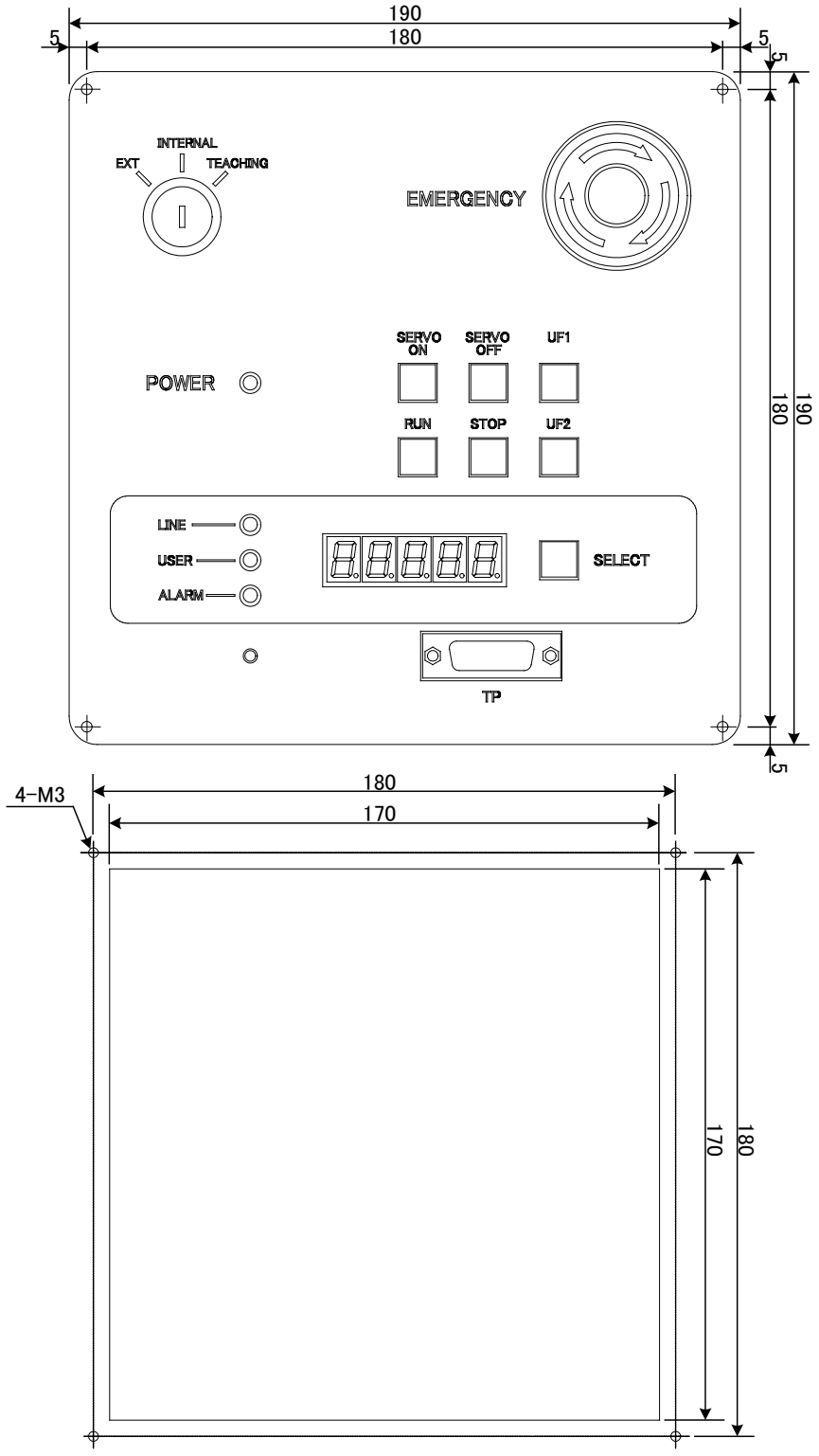


Fig. 3.6 Control panel mounting dimensions

3.3.4 Mounting Dummy Panel on Controller

When the control panel has been disengaged from the controller, mount a dummy panel on the place where the control panel was set before, as shown in Fig. 3.7. The dummy panel, set parts, etc. are provided optionally.

a) Connect the cable connector which was disconnected when separating the controller from the control panel, to the rear side of the dummy panel, then screw both ends of the connector. When mounting the connector, use the cross truss screws (ø3 x 6, ZN3–B).

b) Screw the dummy panel into the controller.

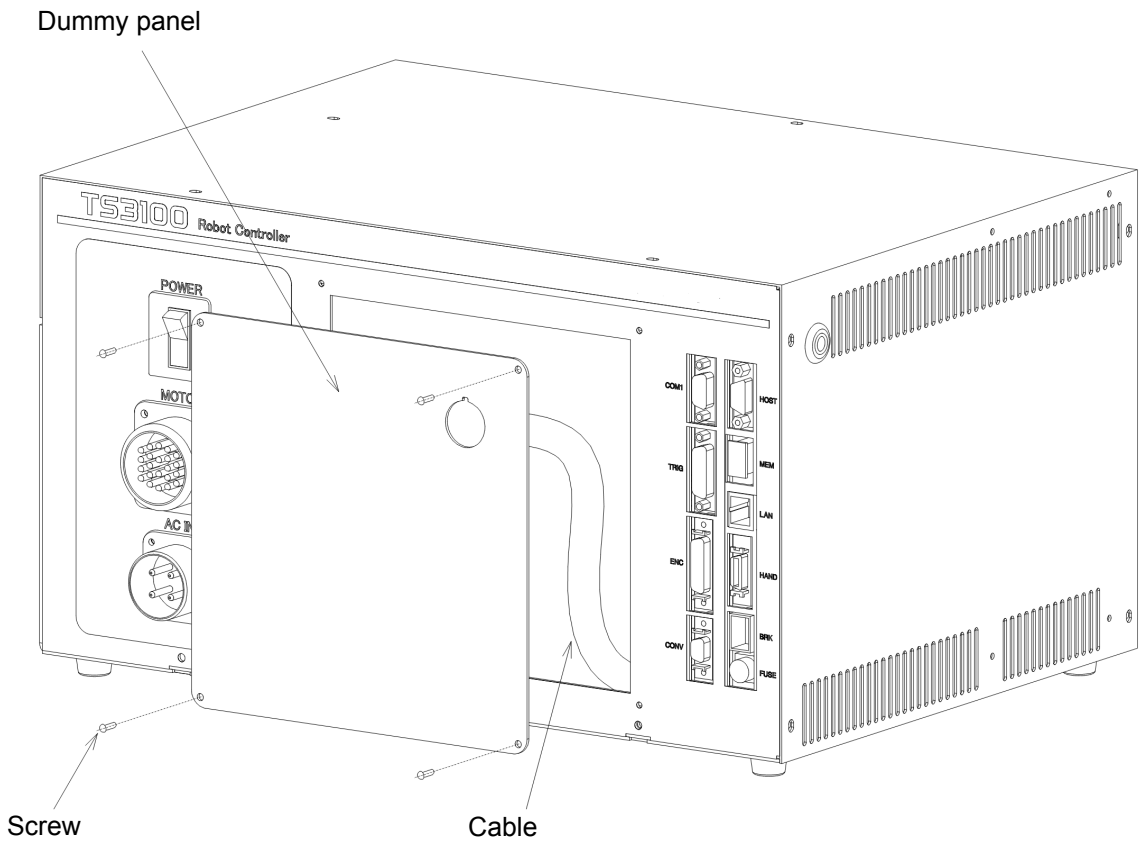


Fig. 3.7 Mounting dummy panel

3.3.5 Dimensions when Separating Control Panel

Fig. 3.8 shows the connections of the control panel and dummy panel. Provide a clearance of 50 mm or more (with cover, 60 mm or more) on the rear side of the separated control panel.
When the cable is connected to the dummy panel of the controller, provide a clearance of 80 mm or so in front of the controller as the cable connector sticks out of the panel surface.

Without cover: 50 mm or more

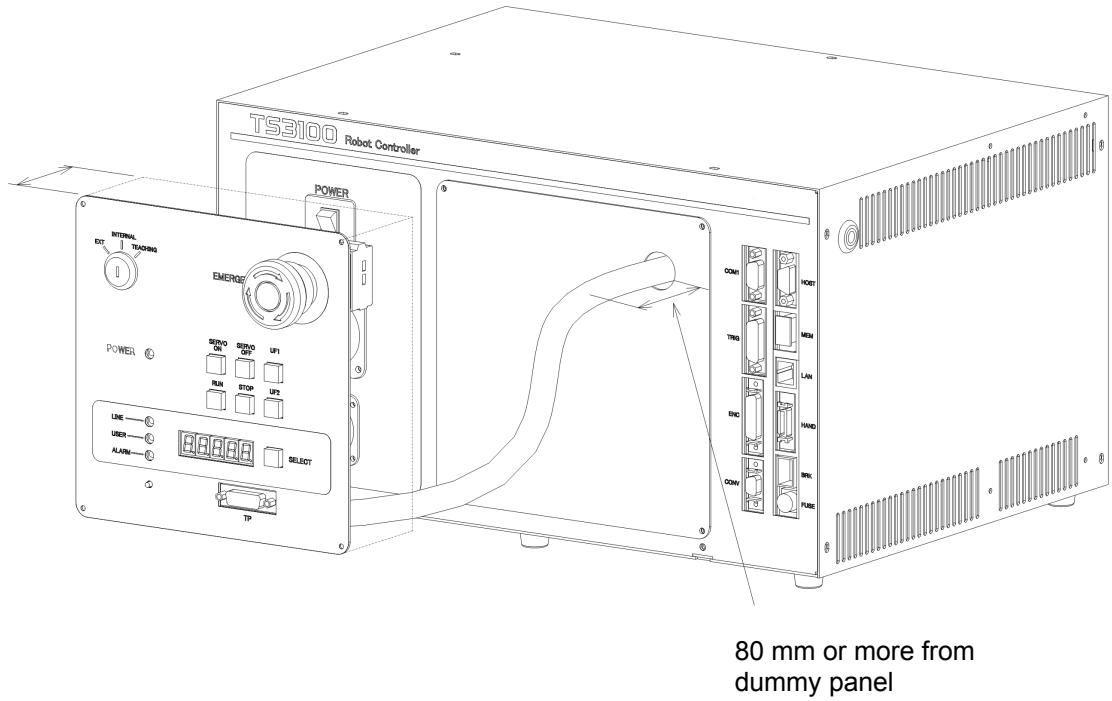


Fig. 3.8 Cable connections of dummy panel and control panel

Section 4 Hand Interface

4.1 Hand Attachment

Dimensions of the hand attachment are shown in Fig. 4.1.

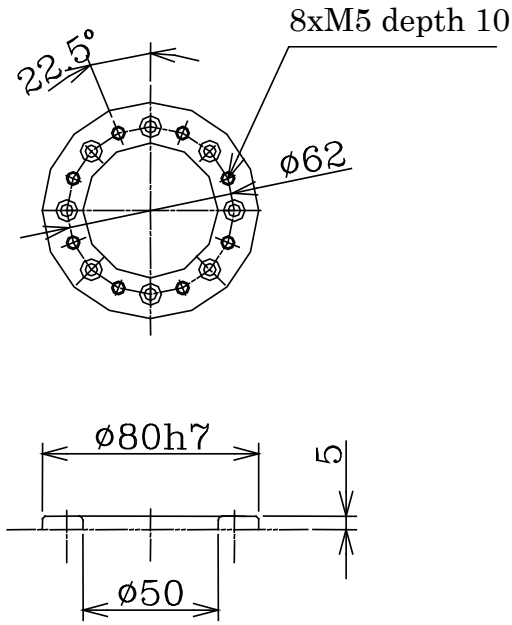


Fig. 4.1 Dimensions of hand attachment

4.2 Wiring and Piping for Hand

For detail of the wiring and the piping for the hand, please refer to the robot specifications.